

ALFIDO TECH INTERNSHIP

Task 3: Collections & Exception Handling

Submitted By: Pravallika Eichapuram

Internship: Java Developer Internship

Task Number: 3

Date: 06 July 2026

Objective :

The objective of this task is to understand Java Collections Framework and Exception Handling. This task demonstrates the use of ArrayList, HashMap, HashSet, try-catch blocks, and custom exception handling.

Programs Implemented :

1. Collections Example :

Description :

This program demonstrates the use of Java Collections such as ArrayList, HashMap, and HashSet to efficiently store and manage data.

Features :

- Uses ArrayList
- Uses HashMap
- Uses HashSet
- Performs data storage and retrieval
- Demonstrates collection operations

2. Exception Handling Example :

Description :

This program demonstrates Java exception handling using try-catch blocks to handle runtime errors safely.

Features :

- Uses try-catch
- Handles runtime exceptions
- Prevents program crashes
- Displays meaningful error message

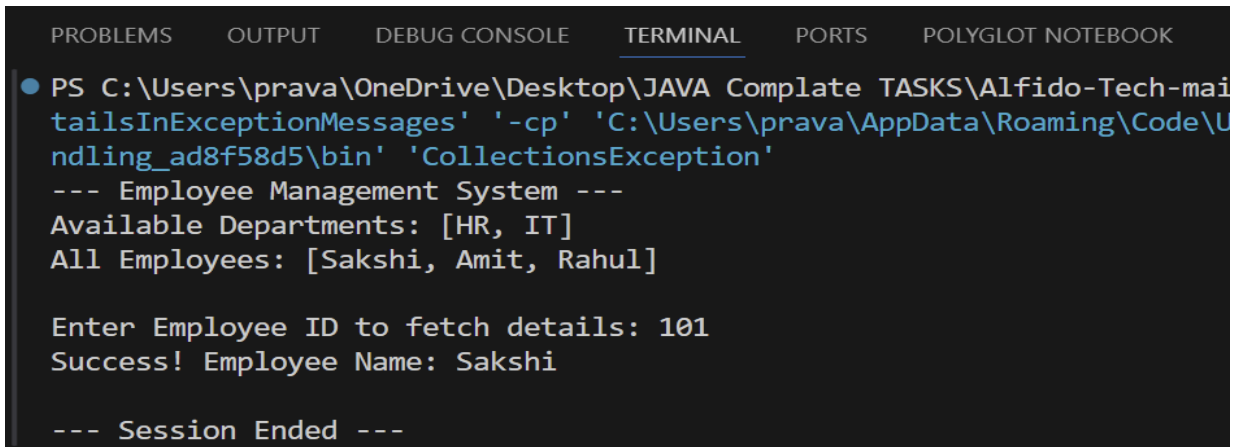
Source Code :

```
J CollectionsException.java •
J CollectionsException.java > CollectionsException > main(String[])
1  import java.util.*;
2
3  // Requirement: Custom Exception (Basic Level)
4  class InvalidEmployeeIDException extends Exception {
5      public InvalidEmployeeIDException(String message) {
6          super(message);
7      }
8  }
9
10 public class CollectionsException {
11     Run | Debug
12     public static void main(String[] args)
13     {
14         // Requirement: Use ArrayList, HashMap, and HashSet
15         List<String> employeeList = new ArrayList<>();
16         Map<Integer, String> employeeMap = new HashMap<>();
17         Set<String> departments = new HashSet<>();
18
19         Scanner sc = new Scanner(System.in);
20
21         try {
22             // 1. ArrayList: Sequence maintain karne ke liye
23             employeeList.add(e: "Sakshi");
24             employeeList.add(e: "Amit");
25             employeeList.add(e: "Rahul");
26
27             // 2. HashMap: ID wise data mapping ke liye
28             employeeMap.put(key: 101, value: "Sakshi");
29             employeeMap.put(key: 102, value: "Amit");
30             employeeMap.put(key: 103, value: "Rahul");
31
32             // 3. HashSet: Unique departments store karne ke liye
33             departments.add(e: "IT");
34             departments.add(e: "HR");
35             departments.add(e: "IT"); // Duplicate
```

```
J CollectionsException.java •
J CollectionsException.java > CollectionsException > main(String[])
10 public class CollectionsException {
11     public static void main(String[] args)
12     {
13         System.out.println(x: "--- Employee Management System ---");
14         System.out.println("Available Departments: " + departments);
15         System.out.println("All Employees: " + employeeList);
16
17         System.out.print(s: "\nEnter Employee ID to fetch details: ");
18         int id = sc.nextInt();
19
20         if (!employeeMap.containsKey(id)) {
21             throw new InvalidEmployeeIDException("Access Denied: Employee ID " + id + " does not exist.");
22         } else {
23             System.out.println("Success! Employee Name: " + employeeMap.get(id));
24         }
25     }
26
27     catch (InvalidEmployeeIDException e)
28     {
29         System.out.println(e.getMessage());
30     }
31
32     catch (InputMismatchException e)
33     {
34         System.out.println(x: "Error: Invalid input format. Please enter a numeric ID.");
35     }
36
37     finally
38     {
39         System.out.println(x: "\n--- Session Ended ---");
40         sc.close();
41     }
42 }
43 }
```

Figure 1: CollectionsException.java Source Code

Output :



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POLYGLOT NOTEBOOK
● PS C:\Users\prava\OneDrive\Desktop\JAVA Complate TASKS\Alfido-Tech-main> java -cp 'C:\Users\prava\AppData\Roaming\Code\U
ndling_ad8f58d5\bin' 'CollectionsException'
--- Employee Management System ---
Available Departments: [HR, IT]
All Employees: [Sakshi, Amit, Rahul]

Enter Employee ID to fetch details: 101
Success! Employee Name: Sakshi

--- Session Ended ---
```

Figure 2: CollectionsException Output

Conclusion :

This task helped me understand Java Collections Framework and Exception Handling, improving my ability to manage data and handle runtime errors effectively.